

Exercise J01

Q1. $\lim_{x \rightarrow 3} \frac{(x^3 + 27) \ln(x - 2)}{(x^2 - 9)} =$

- (A) -8 (B) 8 (C) 9 (D) -9

Ans. C

Sol. -

Q2. $\lim_{x \rightarrow 0} \frac{(4^x - 1)^3}{\sin\left(\frac{x}{p}\right) \ln\left(1 + \frac{x^2}{3}\right)} =$

- (A) $9 p (\log 4)$ (B) $3 p (\log 4)^3$ (C) $12 p (\log 4)^3$ (D) $27 p (\log 4)^2$

Ans. B

Sol. -

Q3. $\lim_{x \rightarrow \infty} \frac{x^3 \cdot \sin \frac{1}{x} + x + 1}{x^2 + x + 1} =$

- (A) 0 (B) $1/2$ (C) 1 (D) none

Ans. C

Sol. -

Q4. $\lim_{n \rightarrow \infty} \frac{-3n + (-1)^n}{4n - (-1)^n}$ is

- (A) $-\frac{3}{4}$ (B) $-\frac{3}{4}$ if n is even ; $\frac{3}{4}$ if n is odd

- (C) not exist if n is even ; $-\frac{3}{4}$ if n is odd

- (D) $+1$ if n is even ; does not exist if n is odd

Ans. A

Sol. -

Q5. $\lim_{x \rightarrow 0^+} \left(1 + \tan^2 \sqrt{x}\right)^{5/x} =$

- (A) e^5 (B) e^2 (C) e (D) none

Ans. A

Sol. -

Q6. The limiting value of $(\cos x)^{1/\sin x}$ as $x \rightarrow 0$ is:

(A) 1 (B) e (C) 0 (D) None of these

Ans. D

Sol. -

Q7. $\lim_{n \rightarrow \infty} \frac{5^{n+1} + 3^n - 2^{2n}}{5^n + 2^n + 3^{2n+3}} =$

(A) 5 (B) 3 (C) 1 (D) zero

Ans. D

Sol. -

Q8. $\lim_{x \rightarrow \pi/2} \frac{\left(1 - \tan \frac{x}{2}\right)(1 - \sin x)}{\left(1 + \tan \frac{x}{2}\right)(\pi - 2x)^3} :$

(A) 1/16 (B) - 1/16 (C) 1/32 (D) - 1/32

Ans. C

Sol. -

Q9. If $f(x) = \sum_{\lambda=1}^n \left(x - \frac{1}{\lambda}\right) \left(x - \frac{1}{\lambda+1}\right)$ then $\lim_{n \rightarrow \infty} f(0)$ is.

(A) 1 (B) -1 (C) 2 (D) None

Ans. A

Sol. -

Q10. The value of $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x) \sin 5x}{x^2 \sin 3x}$ is—

(A) 10/3 (B) 3/10 (C) 6/5 (D) 5/6

Ans. A

Sol. -

Q11. $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 3} \right)^x =$

(A) e^4 (B) e^2 (C) e^3 (D) e

Ans. A

Sol. -

Q12. If $\lim_{x \rightarrow 0} \frac{\log(3+x) - \log(3-x)}{x} = k$, the value of k is -

(A) $-\frac{2}{3}$ (B) 0 (C) $-\frac{1}{3}$ (D) $\frac{2}{3}$

Ans. D

Sol. -

Q13. Let $f(a) = g(a) = k$ and their n^{th} derivatives $f^{(n)}(a)$, $g^{(n)}(a)$ exist and are not equal for some n . Further if $\lim_{x \rightarrow a} \frac{f(a)g(x) - f(x)g(a)}{g(x) - f(x)} = 4$ then the value of k is-

(A) 0 (B) 4 (C) 2 (D) 1

Ans. B

Sol. -

Q14. Let α & β be the distinct roots of $ax^2 + bx + c = 0$, then $\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$ is equal to -

(A) $\frac{a^2}{2}(\alpha - \beta)^2$ (B) 0 (C) $-\frac{a^2}{2}(\alpha - \beta)^2$ (D) $\frac{1}{2}(\alpha - \beta)^2$

Ans. A

Sol. -

Q15. $\lim_{x \rightarrow 0^+} \frac{\ln(-\cos 3x + \cos x)}{\ln \sin x}$ is equal to :

(A) 0 (B) 1 (C) 2 (D) D.N.E.

Ans. C

Sol.
$$\begin{aligned} & \lim_{x \rightarrow 0^+} \frac{\ln(-4\cos^3 x + 3\cos x + \cos x)}{\ln \sin x} \\ & \lim_{x \rightarrow 0^+} \frac{\ln(4\cos x \sin^2 x)}{\ln \sin x} \\ & \lim_{x \rightarrow 0^+} \frac{\ln(4\cos x) + \ln \sin^2 x}{\ln \sin x} \\ & = \lim_{x \rightarrow 0^+} \frac{2 \ln \sin x}{\ln \sin x} + \frac{\ln(4\cos x)}{\ln \sin x} = 2 + 0 = 2 \end{aligned}$$

Q16. The value of $\lim_{x \rightarrow 0} (\sin^{-1}[\sin x] + \cos^{-1}[\cos x] - 2 \tan^{-1}[\tan x])$ is equal to [Note : $[k]$ denotes largest integer function less than or equal to k .]

(A) π (B) $\pi/2$ (C) $3\pi/2$ (D) Non-existent

Ans. B

Sol. -

Q17. Given $l_1 = \lim_{x \rightarrow \frac{\pi}{4}} \cos^{-1} \left[\sec \left(x - \frac{\pi}{4} \right) \right]; \quad l_2 = \lim_{x \rightarrow \frac{\pi}{4}} \sin^{-1} \left[\operatorname{cosec} \left(x + \frac{\pi}{4} \right) \right];$
 $l_3 = \lim_{x \rightarrow \frac{\pi}{4}} \tan^{-1} \left[\cot \left(x + \frac{\pi}{4} \right) \right]; \quad l_4 = \lim_{x \rightarrow \frac{\pi}{4}} \cot^{-1} \left[\tan \left(x - \frac{\pi}{4} \right) \right]$

where $[x]$ denotes greatest integer function then which of the following limits exist ?

(A) l_1 and l_2 only (B) l_1 and l_3 only (C) l_1 and l_4 only (D) All of them

Ans. A

Sol. $l_1 = \lim_{x \rightarrow \frac{\pi}{4}} \cos^{-1} \left[\sec \left(x - \frac{\pi}{4} \right) \right]$

$$\text{LHL} = \lim_{h \rightarrow 0} \cos^{-1} [\sec(-h)] = 0$$

$$\text{RHL} = \lim_{h \rightarrow 0} \cos^{-1} [\sec h] = 0$$

$$l_2 = \lim_{x \rightarrow \frac{\pi}{4}} \sin^{-1} \left[\operatorname{cosec} \left(x + \frac{\pi}{4} \right) \right]$$

$$\text{LHL} = \lim_{h \rightarrow 0} \sin^{-1} \left[\operatorname{cosec} \left(\frac{\pi}{2} - h \right) \right] = \frac{\pi}{2}$$

$$\text{RHL} = \lim_{h \rightarrow 0} \sin^{-1} \left[\operatorname{cosec} \left(\frac{\pi}{2} + h \right) \right]$$

$$= \lim_{h \rightarrow 0} \sin^{-1} [\sec(h)] = \frac{\pi}{2}$$

$$l_3 = \lim_{x \rightarrow \frac{\pi}{4}} \tan^{-1} \left[\cot \left(x + \frac{\pi}{4} \right) \right]$$

$$\text{LHL} = \lim_{h \rightarrow 0} \tan^{-1} \left[\cot \left(\frac{\pi}{4} - h \right) \right] = \lim_{h \rightarrow 0} \tan^{-1} [\tan h] = 0$$

$$\text{RHL} = \lim_{h \rightarrow 0} \tan^{-1} \left[\cot \left(\frac{\pi}{2} + h \right) \right] = \lim_{h \rightarrow 0} \tan^{-1} [-\tan h] = -\frac{\pi}{4}$$

$\Rightarrow l_3$ does not exist

$$l_4 = \lim_{x \rightarrow \frac{\pi}{4}} \cot^{-1} \left[\tan \left(x - \frac{\pi}{4} \right) \right]$$

$$\text{LHL} = \lim_{h \rightarrow 0} \cot^{-1} [-\tan h] = -\frac{\pi}{4}$$

$$\text{RHL} = \lim_{h \rightarrow 0} \cot^{-1} [\tan h] = \cot^{-1}(0) = \frac{\pi}{2}$$

$\Rightarrow l_4$ does not exist

Q18. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{\cos^{-1} \left[\frac{1}{4}(3 \sin x - \sin 3x) \right]}$ where $[]$ denotes greatest integer function, is –

(A) $2/\pi$ (B) 1 (C) $4/\pi$ (D) Does not exist

Ans. A

Sol. -

Q19. The value of $\lim_{x \rightarrow 0} \frac{(\tan(\{x\} - 1)) \sin\{x\}}{\{x\{x\} - 1)}$ where $\{x\}$ denotes the fractional part function :

(A) is 1 (B) is $\tan 1$ (C) is $\sin 1$ (D) is non existent

Ans. D

Sol. -

Q20. $\lim_{x \rightarrow 1} (1 - x + [x - 1] + [1 - x]) =$ where $[x]$ denotes greatest integer function.

(A) 0 (B) 1 (C) -1 (D) dose not exist

Ans. C

Sol. -

Q21. The value of $\lim_{x \rightarrow \pi/4} (1 + [x])^{1/\ln(\tan x)}$ (where $[.]$ denotes the greatest integer function) is equal to

(A) 0 (B) 1 (C) e (D) e^{-1}

Ans. B

Sol. -

- Q22. The limit $\lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right)$, where $[x]$ is the greatest integer function and $n \in \mathbb{N}$, is :
- (A) $2n$ (B) $2n + 1$ (C) $2n - 1$ (D) does not exist

Ans. C

Sol. $\sin \theta < \theta < \tan \theta, \quad \theta \in \left(0, \frac{\pi}{2}\right)$

$$\frac{\sin \theta}{\theta} < 1 < \frac{\tan \theta}{\theta}$$

$$\frac{n \sin \theta}{\theta} < n < \frac{n \tan \theta}{\theta}$$

$$\lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right); n \in \mathbb{N}$$

$$\text{LHL} = \lim_{\theta \rightarrow 0^-} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right) = n - 1 + n = 2n - 1$$

$$\text{RHL} = \lim_{\theta \rightarrow 0^+} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right) = n - 1 + n = 2n - 1$$

$$\therefore \text{LHL} = \text{RHL} = 2n - 1$$

- Q23. The value of $\lim_{x \rightarrow 0} (x^{16} + 4x^{12} - 3x^4) \left[\frac{1}{x^4} \right]$, (where $[.]$ denotes greatest integer function), is –
- (A) 0 (B) -3 (C) 4 (D) Non-existent

Ans. B

Sol. -

- Q24. If $[x]$ denotes the greatest integer $\leq x$, then $\lim_{n \rightarrow \infty} \frac{1}{n^4} ([1^3x] + [2^3x] + \dots + [n^3x])$ equals
- (A) $x/2$ (B) $x/3$ (C) $x/6$ (D) $x/4$

Ans. D

Sol. $\lim_{n \rightarrow \infty} \frac{1}{n^4} \{ [1^3x] + [2^3x] + \dots + [n^3x] \}$

$$1^3x - 1 < [1^3x] \leq 1^3x$$

$$2^3x - 1 < [2^3x] \leq 2^3x$$

$$\vdots \quad \quad \quad \vdots$$

$$n^3x - 1 < [n^3x] \leq n^3x$$

Adding all these inequities

$$(1^3 + 2^3 + 3^3 + \dots + n^3)x - n < [1^3x] + [2^3x] + \dots + [n^3x] \leq (1^3 + 2^3 + \dots + n^3)x$$

$$\frac{n^2(n+1)^2x - n}{n^4} < \frac{|1^3x| + |2^3x| + \dots + |n^3x|}{n^4} \leq \frac{\left(\frac{n(n+1)^2}{2}\right)x}{n^4}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^2 \frac{x}{4} - \frac{1}{n^3} < \lim_{n \rightarrow \infty} \frac{[1^3x] + [2^3x] + \dots + [n^3x]}{n^4} \leq \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^2 \frac{x}{4}$$

$$\frac{x}{4} < \lim_{n \rightarrow \infty} \frac{[1^3x] + [2^3x] + \dots + [n^3x]}{n^4} \leq \frac{x}{4}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{[1^3x] + [2^3x] + \dots + [n^3x]}{n^4} = \frac{x}{4}$$

Q25. $\lim_{x \rightarrow \infty} \frac{\log x^n - [x]}{[x]}$, $n \in \mathbb{N}$, (where $[x]$ denotes greatest integer less than or equal to x)

(A) Has value -1 (B) Has value 0 (C) Has value 1 (D) Does not exist

Ans. D

Sol. -

Q26. Evaluate $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \dots + \frac{1}{\sqrt{n^2+2n}} \right)$

(A) 1 (B) $1/2$ (C) 0 (D) 2

Ans. D

Sol. -

Q27. $\lim_{x \rightarrow \infty} \frac{3}{x} \left[\frac{x}{4} \right] = \frac{p}{q}$ (where $[.]$ denotes greatest integer function), then $p + q$ (where p, q are relative prime) is :

(A) 2 (B) 7 (C) 5 (D) 6

Ans. B

Sol. -

Q28. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a positive increasing function with $\lim_{x \rightarrow \infty} \frac{f(3x)}{f(x)} = 1$. Then $\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} =$

(A) 1 (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 3

Ans. A

Sol. -

Q29. For each $t \in \mathbb{R}$, let $[t]$ be the greatest integer less than or equal to t . Then $\lim_{x \rightarrow 0^+} x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{15}{x} \right] \right)$

(A) is equal to 120 (B) does not exist (in $\mathbb{R}$) (C) is equal to 0 (D) is equal to 15

Ans. A

Sol. $x - 1 < [x] \leq x$

$$\left(\frac{1}{x} - 1\right) < \left[\frac{1}{x}\right] \leq \frac{1}{x}$$

$$x\left(\frac{1}{x} - 1\right) < x\left[\frac{1}{x}\right] \leq \frac{x}{x}$$

use sandwich

$$\lim_{x \rightarrow 0} x\left[\frac{1}{x}\right] = 1$$

$$\text{Similarly } \lim_{x \rightarrow 0} x\left[\frac{2}{x}\right] = 2$$

$\vdots$
 $\vdots$
 $\vdots$

$$\lim_{x \rightarrow 0} x\left[\frac{15}{x}\right] = 15$$

$$\text{Ans} = 1 + 2 + \dots + 15 = 120$$

Q30. If $f(1) = 1$, $f'(1) = 2$, then $\lim_{x \rightarrow 1} \frac{\sqrt{f(x)} - 1}{\sqrt{x} - 1} =$

(A) 2 (B) 1 (C) 3 (D) 4

Ans. A

Sol. -

Q31. If function $f(x)$ is differentiable at $x = a$ then $\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a} =$

(A) $2a f(a) + a^2 f'(a)$ (B) $-a^2 f'(a)$ (C) $af(a) - a^2 f'(a)$ (D) $2af(a) - a^2 f'(a)$

Ans. D

Sol. -

Q32. $\lim_{x \rightarrow 0} \frac{(\cos x) \ln(1+x) - x \cos x - \ln(1+x) + x}{x^4}$ is equal to -

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{8}$

Ans. C

Sol. -

Q33. $\lim_{x \rightarrow 0} \frac{e^{-x^2/2} - \cos x}{x^3 \sin x}$

- (A) 1/4 (B) 1/6 (C) 1/12 (D) 1/8

Ans. C

Sol. -

Q34. If $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2}\right)^{2x} = e^2$, then the values of a and b, are-

- (A) $a \in \mathbb{R}, b \in \mathbb{R}$ (B) $a = 1, b \in \mathbb{R}$ (C) $a \in \mathbb{R}, b = 2$ (D) $a = 1$ and $b = 2$

Ans. B

Sol. -

Q35. The value of $\lim_{x \rightarrow 0} \frac{\cos(\sin x) - \cos x}{x^4}$ is equal to:

- (A) 1/5 (B) 1/6 (C) 1/4 (D) 1/2

Ans. B

Sol.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\cos(\sin x) - \cos x}{x^4} &= \lim_{x \rightarrow 0} \frac{\cos\left(x - \frac{x^3}{3!} + \dots\right) - \cos x}{x^4} \\ &= \lim_{x \rightarrow 0} \frac{\left[\frac{1 - \left(x - \frac{x^3}{6} + \dots\right)^2}{2!} + \frac{\left(x - \frac{x^3}{6} + \dots\right)^4}{4!} - \dots \right] - \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right]}{x^4} \\ &= \lim_{x \rightarrow 0} \frac{x^4 \left(\frac{1}{2} \cdot \frac{2}{6} + \frac{1}{24} \right) - \frac{x^4}{24} + \dots}{x^4} = \frac{1}{6} + \frac{1}{24} - \frac{1}{24} = \frac{1}{6} \end{aligned}$$

Q36. The integer 'n' for which $\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite non-zero number is:

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. C

Sol. -

Exercise J02

Q1. $\lim_{x \rightarrow 1} \frac{x^2 - x \cdot \ln x + \ln x - 1}{x - 1}$

Ans. 2

Sol. -

Q2. $\lim_{x \rightarrow 1} \frac{\left[\sum_{k=1}^{100} X^k \right] - 100}{x - 1}$

Ans. 5050

Sol. -

Q3. $\lim_{x \rightarrow 1} \frac{1 - \tan x}{1 - \sqrt{2} \sin x}$

Ans. 2

Sol. -

Q4. Find the sum of an infinite geometric series whose first term is the limit of the function $f(x) = \frac{\tan x - \sin x}{\sin^3 x}$ as $x \rightarrow 0$ and whose common ratio is the limit of the function $g(x) = \frac{1 - \sqrt{x}}{(\cos^{-1} x)^2}$ as $x \rightarrow 1$.

Ans. $a = \frac{1}{2}; r = \frac{1}{4} = \frac{1}{4}; S = \frac{2}{3}$

Sol. -

Q5. $\lim_{x \rightarrow \infty} (x - \ln \cosh x)$ where $\cosh t = \frac{e^t + e^{-t}}{2}$.

Ans. $\ln 2$

Sol. -

Q6. $\lim_{x \rightarrow \frac{3\pi}{4}} \frac{1 + \sqrt[3]{\tan x}}{1 - 2 \cos^2 x}$

Ans. $-\frac{1}{3}$

Sol. -

Q7. $\lim_{x \rightarrow 0} \left[1 - \cos \frac{x^2}{2} - \cos \frac{x^2}{4} + \cos \frac{x^2}{2} \cos \frac{x^2}{4} \right]$

Ans. $\frac{1}{32}$

Sol. -

Q8. If $l = \lim_{x \rightarrow 0} \sum_{r=2}^n \left((r+1) \sin \frac{\pi}{r+1} - r \sin \frac{\pi}{r} \right)$ then find $\{l\}$. (where $\{ \}$ denotes the fractional part function)

Ans. $\pi - 3$

Sol. -

Q9. $\lim_{x \rightarrow 1} \frac{(\ln(1+x) - \ln 2(3.4^{x-1} - 3x))}{[(7+x)^{\frac{1}{3}} - (1+3x)^{\frac{1}{2}}] \cdot \sin(x-1)}$

Ans. $-\frac{9}{4} \ln \frac{4}{e}$

Sol. -

Q10. $\lim_{x \rightarrow 1} \frac{27^x - 9^x - 3^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}}$

Ans. $8\sqrt{2}(\ln 3)^2$

Sol. -

Q11. Let $P_n = a^{P_{n-1}} - 1, \forall n = 2, 3, \dots$ and Let $P_1 = a^x - 1$ where $a \in \mathbb{R}^+$ then evaluate

$\lim_{x \rightarrow 0} \frac{P_n}{x}$.

Ans. $(\ln a)n$

Sol. -

Q12. $\lim_{y \rightarrow 0} \left[\frac{\exp\left(x \ln\left(1 + \frac{ay}{x}\right)\right) - \exp\left(x \ln\left(1 + \frac{by}{x}\right)\right)}{y} \right]$

Ans. $a - b$

Sol. -

Q13. Let $\{a_n\}, \{b_n\}, \{c_n\}$ be sequences such that

(i) $a_n + b_n + c_n = 2n + 1$; (ii) $a_n b_n + b_n c_n + c_n a_n = 2n - 1$; (iii) $a_n b_n c_n = -1$; (iv) $a_n < b_n < c_n$

Then find the value of $\lim_{n \rightarrow \infty} (na_n)$.

Ans. $-1/2$

Sol. -

Q14. $\lim_{x \rightarrow 0} \left(\frac{x - 1 + \cos x}{x} \right)^{\frac{1}{x}}$

Ans. $e^{-1/2}$

Sol. -

Q15. If $n \in \mathbb{N}$ and $a_n = 2^2 + 4^2 + 6^2 + \dots + (2n)^2$ and $b_n = 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2$.
Find the value $\lim_{n \rightarrow \infty} \frac{\sqrt{a_n} - \sqrt{b_n}}{\sqrt{n}}$.

Ans. $\sqrt{3}/2$

Sol. -

Q16. If $|x| < 1$, then $\lim_{n \rightarrow \infty} ((1+x)(1+x^2)\dots(1+x^{2^n}))$ equal to :

Ans. $\frac{1}{1-x}$

Sol. Given $|x| < 1$

$$\begin{aligned} & \lim_{n \rightarrow \infty} ((1+x)(1+x^2)\dots(1+x^{2^n})) \\ &= \lim_{n \rightarrow \infty} 1 + x + x^2 + \dots + x^{2^{n+1}} \\ &= \frac{1}{1-x} \end{aligned}$$

Q17. Let $f(x) = \begin{cases} -5 & , \quad -\infty < x \leq -1 \\ x-4 & , \quad -1 < x \leq 6 \\ 2 & , \quad 6 < x < \infty \end{cases}$ then $\lim_{x \rightarrow n} \frac{[x]^2 - 13[x] + 42}{(x-7)(x-6)}$ is equal to (where 'n' is the number of non zero integers in the range of f(x), [.] denotes greatest integer function)

Ans. 0

Sol. -

Q18. The value of $\lim_{x \rightarrow 0} \left[\frac{\sin^{-1} x}{x} \right] + 5 \left[\frac{\sin^{-1} 2x}{x} \right] + 20 \left[\frac{x}{\tan^{-1} x} \right] + 10 \left[\frac{x}{\tan^{-1} 2x} \right]$ (where [.] denotes greatest integer function) is equal to –

Ans. 20

Sol. -

Q19. (a) $\lim_{x \rightarrow \frac{1}{\sqrt{2}}} \frac{\cos^{-1} \left(2x\sqrt{1-x^2} \right)}{x - \frac{1}{\sqrt{2}}}$; (b) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{1 - \sqrt{\sin 2x}}}{\pi - 4x}$ (c) $\lim_{x \rightarrow -7} \frac{[x]^2 + 15[x] + 56}{\sin(x+7)\sin(x+8)}$ Where [.] denotes the greatest integer function

Ans. (a) DNE (b) DNE (c) 0

Sol. -

Q20. $\lim_{x \rightarrow -\infty} \frac{(3x^4 + 2x^2) \sin \frac{1}{x} + |x|^3 + 5}{|x|^3 + |x|^2 + |x| + 1}$

Ans. -2

Sol. -

Q21. Let $f(x) = \frac{x}{\sin x}$, $x > 0$ and $g(x) = x + 3$, $x < 1$
 $= 2 - x$, $x \leq 0$ $= x^2 - 2x - 2$, $1 \leq x < 2$
 $= x - 5$, $x \geq 2$

find LHL and RHL of at $x = 0$ and hence find $\lim_{x \rightarrow 0} g(f(x))$.

Ans. $-3, -3, -3$

Sol. -

Q22. (a) $\lim_{x \rightarrow 0} \tan^{-1} \frac{a}{x^2}$, where $a \in R$;

(b) Plot the graph of the function $f(x) = \lim_{x \rightarrow 0} \left(\frac{2x}{\pi} \tan^{-1} \frac{x}{t^2} \right)$

Ans. (a) $\pi/2$ if $a > 0$; 0 if $a = 0$ and $-\pi/2$ if $a < 0$; (b) $f(x) = |x|$

Sol. -

Q23. Find the value of $\lim_{n \rightarrow \infty} \left[\frac{2}{\pi} (n+1) \cos^{-1} \left(\frac{1}{n} \right) - n \right]$.

Ans. $1 - \frac{2}{\pi}$

Sol. -

Q24. Using Sandwich theorem, evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{1+n^2} + \frac{2}{2+n^2} + \dots + \frac{n}{n+n^2}$$

Ans. $1/2$

Sol. -

Q25. $\lim_{h \rightarrow 0} \frac{\tan(a+2h) - 2\tan(a+h) + \tan a}{h^2}$

Ans. $2 \sec^2 a \cdot \tan a$

Sol. -

Q26. $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{3} + 4h\right) - 4\sin\left(\frac{\pi}{3} + 3h\right) + 6\sin\left(\frac{\pi}{3} + 2h\right) - 4\sin\left(\frac{\pi}{3} + h\right) + \sin\frac{\pi}{3}}{h^4}$

Ans. $\frac{\sqrt{3}}{2}$

Sol. -

Q27. Let $l_1 = \lim_{x \rightarrow 0} \frac{\sin(\sin^{-1} x) - \sin x}{x^3}$, $l_2 = \lim_{x \rightarrow 0} \frac{\tan(\tan^{-1} x) - \ln(1+x)}{x^2}$, $l_3 = \lim_{x \rightarrow 0} \frac{e^x - e^{\ln x} - 1}{x^2}$
& $l_4 = \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x^3}$

$$l_5 = \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2}{x^4}. \text{ Find the value of } 3(l_1 + 2l_2 + 3l_3 + l_4 \div l_5)$$

Ans. 20

Sol. -

$$\text{Q28. } \lim_{x \rightarrow 1} \left(\frac{p}{1-x^p} - \frac{q}{1-x^q} \right), p, q \in N$$

$$\text{Ans. } \frac{p-q}{2}$$

Sol. -

$$\text{Q29. } \lim_{x \rightarrow 1} x^2 \left(\sqrt{\frac{x+2}{x}} - \sqrt{\frac{x+3}{x}} \right)$$

$$\text{Ans. } 1/2$$

Sol. -

$$\text{Q30. Find a \& b if : (i) } \lim_{x \rightarrow \infty} \left[\frac{x^2+1}{x+1} - ax - b \right] = 0 \quad \text{(ii) } \lim_{x \rightarrow -\infty} \left[\sqrt{x^2-x+1} - ax - b \right] = 0$$

$$\text{Ans. (i) } a=1, b=-1 \quad \text{(ii) } a=-1, b=\frac{1}{2}$$

Sol. -

$$\text{Q31. If the } \lim_{x \rightarrow 0} \frac{1}{x^3} \left(\frac{1}{\sqrt{1+x}} - \frac{1+ax}{1+bx} \right) \text{ exists and has the value equal to } l, \text{ then find the value of } \frac{1}{a} - \frac{2}{l} + \frac{3}{b}.$$

$$\text{Ans. } 72$$

Sol. -

$$\text{Q32. Let } f(x) = ax^3 + bx^2 + cx + d \text{ and } g(x) = x^2 + x - 2.$$

$$\text{If } \lim_{x \rightarrow 1} \frac{f(x)}{g(x)} = 1 \text{ and } \lim_{x \rightarrow -2} \frac{f(x)}{g(x)} = 4, \text{ then find the value of } \frac{c^2 + d^2}{a^2 + b^2}.$$

$$\text{Ans. } 16$$

Sol. -

$$\text{Q33. Find the values of constants a, b \& c so that } \lim_{x \rightarrow 0} \frac{axe^x - b \log(1+x) + cxe^{-x}}{x^2 \sin x} = 2.$$

$$\text{Ans. } a=3, b=12, c=9$$

Sol. -

$$\text{Q34. If } \lim_{x \rightarrow \infty} \frac{a(2x^3 - x^2) + b(x^3 + 5x^2 - 1) - c(3x^3 + x^2)}{a(5x^4 - x) - bx^4 + c(4x^4 + 1) + 2x^2 + 5x} = 1, \text{ then the value of } (a+b+c) \text{ can be expressed in the lowest form as } \frac{p}{q}. \text{ Find the value of } (p+q).$$

$$\text{Ans. } 167$$

Sol. -

Q35. $f(x)$ is the function such that $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$. If $\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{(f(x))^3} = 1$, then find the value of a and b .

Ans. $a = -5/2, b = -3/2$

Sol. -

Q36. If $\lim_{x \rightarrow 1} (1 + ax + bx^2)^{\frac{c}{x-1}} = e^3$, then the value of $\left| \frac{5b}{a} \right| + 4bc$ is equal to.

Ans. 17

Sol. $\lim_{x \rightarrow 1} (1 + ax + bx^2)^{\frac{c}{x-1}} = e^3$

$$\Rightarrow 1 + a + b = 1$$

$$\Rightarrow \exp \left(\lim_{x \rightarrow 1} \frac{acx + bcx^2}{x-1} \right) = e^3$$

$$\Rightarrow (a + b)c = 0$$

$$\Rightarrow \lim_{x \rightarrow 1} c \left(\frac{ax + bx^2}{x-1} \right) = 3$$

$$\Rightarrow c \lim_{x \rightarrow 1} \frac{a(x-1+1) + b(x-1)(x+1) + b}{x-1} = 3$$

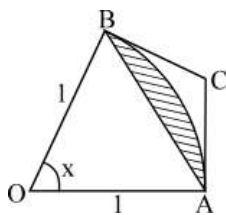
$$\Rightarrow c \lim_{x \rightarrow 1} \left(a + b(x+1) + \frac{a+b}{x-1} \right) = 3$$

$$\Rightarrow (a + 2b)c = 3$$

$$\Rightarrow bc = 3$$

$$\Rightarrow \left| \frac{5b}{a} \right| + 4bc = 17$$

Q37. A circular arc of radius 1 subtends an angle of x radians, $0 < x < \frac{\pi}{2}$ as shown in the figure. The point C is the intersection of the two tangent lines at A & B . Let $T(x)$ be the area of triangle ABC & let $S(x)$ be the area of the shaded region. Compute:



(a) $T(x)$ (b) $S(x)$ (c) the limit of $\frac{T(x)}{S(x)}$ as $x \rightarrow 0$.

Ans. $T(x) = \frac{1}{2} \tan^2 \frac{x}{2} \cdot \sin x$ or $\tan \frac{x}{2} - \frac{\sin x}{2}$, $S(x) = \frac{1}{2}x - \frac{1}{2} \sin x$, limit = $\frac{3}{2}$

Sol. -

Paragraph

Consider two functions $f(x) = \lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} \right)^n$ and $g(x) = -x^{4b}$ where $b = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1} \right)$ Then

\n

Q38. $f(x)$ is

- (A) e^{-x^2} (B) $e^{-\frac{x^2}{2}}$ (C) e^{x^2} (D) $e^{\frac{x^2}{2}}$

Ans. B

Sol. -

Paragraph

Consider two functions $f(x) = \lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} \right)^n$ and $g(x) = -x^{4b}$ where $b = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1} \right)$ Then

\n

Q39. $g(x)$ is

- (A) $-x^2$ (B) x^2 (C) x^4 (D) $-x^4$

Ans. A

Sol. -

Paragraph

Consider two functions $f(x) = \lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} \right)^n$ and $g(x) = -x^{4b}$ where $b = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1} \right)$ Then

\n

Q40. Number of solutions of $f(x) + g(x) = 0$ is

- (A) 2 (B) 4 (C) 0 (D) 1

Ans. A

Sol. -

Paragraph

Consider $f(x) = \frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^3}$, where a, b, c are real numbers.

\n

Q41. If $\lim_{x \rightarrow 0^+} f(x)$ is finite, then the value of $a + b + c$ is

- (A) 0 (B) 1 (C) 2 (D) -2

Ans. A

Sol. -

Paragraph

Consider $f(x) = \frac{\sin x + ae^x + be^{-x} + c\lambda n(1+x)}{x^3}$, where a, b, c are real numbers.

\n

Q42. If $f(x) = \ell$ (finite), then the value of ℓ is

- (A) -2 (B) $-\frac{1}{2}$ (C) -1 (D) $-\frac{1}{3}$

Ans. D

Sol. -

Paragraph

Consider $f(x) = \frac{\sin x + ae^x + be^{-x} + c\lambda n(1+x)}{x^3}$, where a, b, c are real numbers.

\n

Q43. Using the values of a, b, c as found above, the value of $\lim_{x \rightarrow 0^+} x f(x)$ is

- (A) 0 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) 2

Ans. A

Sol. -

Paragraph

Let $f(x) = \begin{cases} x+3; & -2 < x < 0 \\ 4 & ; x = 0 \\ 2x+5; & 0 < x < 1 \end{cases}$, then :

\n

Q44. $\lim_{x \rightarrow 0^+} \left(\left\{ \frac{x}{\tan x} \right\} \right)$ is : (where $\{.\}$ denotes fractional part of function)

- (A) 4 (B) 5 (C) 7 (D) None of these

Ans. C

Sol. -

Paragraph

Let $f(x) = \begin{cases} x+3; & -2 < x < 0 \\ 4 & ; x=0 \\ 2x+5; & 0 < x < 1 \end{cases}$, then :

\n

Q45. $\lim_{x \rightarrow 0^-} f([x - \tan x])$ is : (where $[.]$ denotes greatest integer function)

- (A) 2 (B) 4 (C) 5 (D) None of these

Ans. B

Sol. -

Q46. If $\lim_{x \rightarrow 0} \frac{\sin 2x + a \sin x}{x^3} = p$ (finite), then

- (A) $a = -2$ (B) $a = -1$ (C) $p = -2$ (D) $p = -1$

Ans. A, D

Sol. -

Q47. Let $f(x) = \frac{|x + \pi|}{\sin x}$, then

- (A) $f(-\pi^+) = -1$ (B) $f(-\pi^-) = 1$ (C) $\lim_{x \rightarrow -\pi} f(x)$ does not exist

- (D) $\lim_{x \rightarrow \pi} f(x)$ does not exist

Ans. D, C, B, A

Sol. -

Q48. $\lim_{x \rightarrow \infty} \frac{(ax + 1)^n}{x^n + A}$

- (A) a^n if $n \in \mathbb{N}$ (B) ∞ if $n \in \mathbb{Z}^-$ & $a = A = 0$ (C) $\frac{1}{1+A}$ if $n = 0$

- (D) a^n if $n \in \mathbb{Z}^-, A = 0$ & $a \neq 0$

Ans. A, B, C, D

Sol. -

Q49. Let $f(x) = \frac{\sqrt{x^2 + 2}}{3x - 6}$ =

- (A) $\lim_{x \rightarrow -\infty} f(x) = -\frac{1}{3}$ (B) $\lim_{x \rightarrow \infty} f(x) = \frac{1}{3}$ (C) $\lim_{x \rightarrow -\infty} f(x) = \frac{1}{3}$ (D) $\lim_{x \rightarrow \infty} f(x) = -\frac{1}{3}$

Ans. A, B

Sol. -

Q50. If $\lambda = \lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{1 + a \cos x}{x^2} - \lim_{x \rightarrow 0} \frac{b \sin x}{x^3}$, where $\lambda \in \mathbb{R}$, then

- (A) $(a, b) = (-1, 0)$ (B) a & b are any real numbers (C) $\lambda = 0$ (D) $\lambda = \frac{1}{2}$

Ans. A, D

Sol. -

Q51. $\lim_{x \rightarrow c} f(x)$ does not exist when

where $[x]$ denotes step up function & $\{x\}$ fractional part function.

- (A) $f(x) = [[x]] - [2x - 1]$, $c = 3$ (B) $f(x) = [x] - x$, $c = 1$
 (C) $f(x) = \{x\}^2 - \{-x\}^2$, $c = 0$ (D) $f(x) = \frac{\tan(\operatorname{sgn} x)}{\operatorname{sgn} x}$, $c = 0$

Ans. B, C

Sol. -

Q52. Which of the following limits vanish?

where $[]$ denotes greatest integer function

- (A) $\lim_{x \rightarrow \infty} x^{\frac{1}{4}} \sin \frac{1}{\sqrt{x}}$ (B) $\lim_{x \rightarrow \infty} (1 - \sin x) \cdot \tan x$ (C) $\lim_{x \rightarrow \infty} \frac{2x^2 + 3}{x^2 + x - 5} \cdot \operatorname{sgn}(x)$
 (D) $\lim_{x \rightarrow 3^+} \frac{[x]^2 - 9}{x^2 - 9}$

Ans. A, B, D

Sol. -

Q53. If $\lim_{x \rightarrow 0} (p \tan qx^2 - 3 \cos^2 x + 4)^{1/(3x^2)} = e^{5/3}$; $p, q \in \mathbb{R}$ then :

- (A) $p = \sqrt{2}, q = \frac{1}{2\sqrt{2}}$ (B) $p = \frac{1}{\sqrt{2}}, q = 2\sqrt{2}$ (C) $p = 1, q = 2$ (D) $p = 2, q = 4$

Ans. C, B

Sol. -

Q54. Let $f: \mathbb{R} \rightarrow [-1, 1]$ be defined as $f(x) = \cos(\sin x)$, then which of the following is/are correct :

- (A) f is periodic with fundamental period 2π (B) Range of $f = [\cos 1, 1]$
 (C) $\lim_{x \rightarrow \frac{\pi}{2}} \left(f\left(\frac{\pi}{2} - x\right) + f\left(\frac{\pi}{2} + x\right) \right) = 2$ (D) f is neither even nor odd function

Ans. B, C

Sol. -

Q55. If $\lim_{x \rightarrow 0} \frac{(\sin(\sin x) - \sin x)}{ax^3 + bx^5 + c} = \frac{1}{12}$, then :

- (A) $a = 2$ (B) $a = -2$ (C) $c = 0$ (D) $b \in \mathbb{R}$

Ans. D, C, A

Sol. -

Q56. $\lim_{x \rightarrow 0} f(x)$, where $f(x)$ is as in **column - I** is

Column – I

(A) $f(x) = \frac{\tan[e^2]x^2 - \tan[-e^2]x^2}{\sin^2 x}$

where $[x]$ is the greatest integer function.

(B) $f(x) = \frac{[5/2 + \tan^2 x + \tan^2 x] - [5/2]}{\tan x}$

where $[x]$ is the greatest integer function.

(C) $f(x) = \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$

(D) $f(x) = \frac{\sqrt{2} - \sqrt{1 - \cos x}}{\sin^2 x}$

Column – II

(P) $\sqrt{2}/8$

(Q) 15

(R) 0

(S) $1/2$

(A) (A) $\rightarrow$ Q (B) $\rightarrow$ R (C) $\rightarrow$ S (D) $\rightarrow$ P

(B) (A) $\rightarrow$ P (B) $\rightarrow$ R (C) $\rightarrow$ S (D) $\rightarrow$ Q

(C) (A) $\rightarrow$ Q (B) $\rightarrow$ S (C) $\rightarrow$ R (D) $\rightarrow$ P

(D) (A) $\rightarrow$ R (B) $\rightarrow$ Q (C) $\rightarrow$ P (D) $\rightarrow$ S

Ans. 5f13eb9190e4b0a1523ca8ea

Sol. -

Q57. Column-I

Column-II

(A) $\lim_{x \rightarrow \infty} \left(\sqrt{x + \sqrt{x}} - \sqrt{x - \sqrt{x}} \right)$ equals

(P) -2

(B) The value of the limit, $\lim_{x \rightarrow 0} \frac{\sin 2x - 2 \tan x}{\ln(1+x^3)}$ is

(Q) -1

(C) $\lim_{x \rightarrow 0^+} \left(\ln \sin^3 x - \ln(x^4 + ex^3) \right)$ equals

(R) 0

(D) Let $\tan(2\pi | \sin \theta |) = \cot(2\pi | \cos \theta |)$, where $\theta \in \mathbb{R}$ and $f(x) = (| \sin \theta | + | \cos \theta |)^x$. The value of $\lim_{x \rightarrow \infty} \left[\frac{2}{f(x)} \right]$ equals

(S) 1

(Here $[]$ represents greatest integer function)

(A) (A) $\rightarrow$ P (B) $\rightarrow$ S (C) $\rightarrow$ Q (D) $\rightarrow$ R (B) (A) $\rightarrow$ S (B) $\rightarrow$ P (C) $\rightarrow$ Q (D) $\rightarrow$ R

(C) (A) $\rightarrow$ S (B) $\rightarrow$ P (C) $\rightarrow$ R (D) $\rightarrow$ Q (D) (A) $\rightarrow$ R (B) $\rightarrow$ Q (C) $\rightarrow$ P (D) $\rightarrow$ S

Ans. 5f13f14190e4b0a1523cb189

Sol. -

Q58. **Column-I** **Column-II**

- | | |
|--|-------------------|
| (A) $\lim_{x \rightarrow 1} \frac{\ln x}{x^4 - 1}$ | (P) 2 |
| (B) $\lim_{x \rightarrow 0} \frac{3e^x - x^3 - 3x}{\tan^2 x}$ | (Q) $\frac{1}{2}$ |
| (C) $\lim_{x \rightarrow \infty} \frac{\pi - 2 \tan^{-1} x}{\ln \left(1 + \frac{1}{x} \right)}$ | (R) $\frac{3}{2}$ |
| (D) $\lim_{x \rightarrow 0} \frac{2 \sin x - \sin 2x}{x(\cos x - \cos 2x)}$ | (S) $\frac{1}{4}$ |
| (E) $\lim_{x \rightarrow 0} \frac{e^x e^{-x} - 2x}{x - \sin x}$ | |

(A) (A) $\rightarrow$ S (B) $\rightarrow$ R (C) $\rightarrow$ P (D) $\rightarrow$ Q (E) $\rightarrow$ P

(B) (A) $\rightarrow$ S (B) $\rightarrow$ R (C) $\rightarrow$ Q (D) $\rightarrow$ P (E) $\rightarrow$ Q

(C) (A) $\rightarrow$ R (B) $\rightarrow$ P (C) $\rightarrow$ P (D) $\rightarrow$ Q (E) $\rightarrow$ R

(D) (A) $\rightarrow$ S (B) $\rightarrow$ P (C) $\rightarrow$ R (D) $\rightarrow$ S (E) $\rightarrow$ Q

Ans. 5f14001f90e4b0a1523ccffb

Sol. -